UNIVERSITY OF GRAZ
Department of Mathematics
and Scientific Computing

MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/2026

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PROJECTS

Due: 03 July 2026, 18:00

1 Herbivor-plant interaction model

In the context of the ODE framework, let consider the following evolution of the plant $p = p(t)$ and of the herbivor $h = h(t)$, $t \geq 0$,

$$\begin{cases} \dot{p} = rp\left(1 - \frac{p}{K}\right) - \sigma\frac{p}{1+p}h, \\ \dot{h} = \gamma\sigma\frac{p}{1+p}h - \mu h, \end{cases} \tag{1}$$

where all the coefficients are strictly positive. Discuss the bifurcation analysis of the model.

(Optional) Include in the analysis the spatial effect of the linear diffusion for both species.

2 Activator-Inhibitor dynamics

We consider the following model, in the context of the activator-inhibitor type dynamics. Interpret its dynamics and perform the Turing instability analysis.

Let consider $u = u(t,x), v = v(t,x)$ with $t \in (0,+\infty)$ and $x \in \Omega \in \mathbb{R}^N$, $N \geq 1$, bounded and regular open set,

$$\begin{cases} \partial_t u - d_u\Delta u = k_1 - k_2 u + k_3\frac{u^2}{v}, & (t,x) \in (0,+\infty)\times\Omega, \\ \partial_t v - d_v\Delta v = k_4 u^2 - k_5 v, & (t,x) \in (0,+\infty)\times\Omega, \end{cases} \tag{2}$$

and the system is endowed with the homogeneous boundary conditions.

1) Determine (if any) the condition such that the Turing instability occurs.

2) How does the scenario change if $f(v) = \frac{1}{v}$ is replaced by $f(u,v) = \dfrac{1}{v(1+ku^2)}$? Explain from both the stability and the modelling viewpoints.

3 A general reaction diffusion system

For the following system, interpret its terms in the context of population dynamics and perform the linear stability analysis.

Let consider $u = u(t,x), v = v(t,x)$ with $t \in (0,+\infty)$ and $x \in \Omega \in \mathbb{R}^N$, $N \geq 1$, bounded and regular open set,

$$\begin{cases} \partial_t u - d_u\Delta u = g(u)u - bf(u)v, & (t,x) \in (0,+\infty)\times\Omega, \\ \partial_t v - d_v\Delta v = bcf(u)v - mv, & (t,x) \in (0,+\infty)\times\Omega, \end{cases} \tag{3}$$

where all the coefficients d_u, d_v, b, c, m are strictly positive. We endow the system with homogeneous boundary conditions.

1) Determine (if any) the condition on the functions f and g such that the Turing instability occurs.

2) How does the scenario change if $f = f(u, v)$? Explain either from the stability and the modelling viewpoint.

4 The Klausmeier Model

Let consider the following reaction diffusion system in $1D$, describing the evolution of the density of the water $w = w(t, x)$ and of the plant density $p = p(t, x)$,

$$\begin{cases} \partial_t w - d_p \dfrac{\partial^2 w}{\partial^2 x} = a - w - wp^2, & (t, x) \in (0, +\infty) \times (-l, l), \\ \partial_t p - d_p \dfrac{\partial^2 p}{\partial^2 x} = wp^2 - bp, & (t, x) \in (0, +\infty) \times (-l, l), \end{cases} \tag{4}$$

where all the coefficients are strictly positive. We endow the system of homogeneous boundary conditions. Perform the Turing instability.

5 Lotka-Volterra interactions and linear diffusion

For the following system, interpret its terms in the context of population dynamics and perform the linear stability analysis. Thus, provide the conditions (if any) under which Turing instability occurs.

Let consider $u = u(t, x), v = v(t, x)$ with $t \in (0, +\infty)$ and $x \in \Omega \in \mathbb{N}$, $N \geq 1$, bounded and regular open set,

$$\begin{cases} \partial_t u - d_u \Delta u = u(r_u - r_{11}u - r_{12}v), & (t, x) \in (0, +\infty) \times \Omega, \\ \partial_t v - d_v \Delta v = v(r_v - r_{21}u - r_{22}v), & (t, x) \in (0, +\infty) \times \Omega, \end{cases} \tag{5}$$

where all the coefficients are strictly positive. We endow the system with homogeneous boundary conditions.

6 Lotka-Volterra interactions and nonlinear diffusion

For the following system, interpret its terms in the context of population dynamics and perform the linear stability analysis. Thus, provide the conditions (if any) under which Turing instability occurs.

Let consider $u = u(t, x), v = v(t, x)$ with $t \in (0, +\infty)$ and $x \in \Omega \in \mathbb{N}$, $N \geq 1$, bounded and regular open set,

$$\begin{cases} \partial_t u - \Delta \left(u(d_u + d_{12}v) \right) = u(r_u - r_{11}u - r_{12}v), & (t, x) \in (0, +\infty) \times \Omega, \\ \partial_t v - d_v \Delta v = v(r_v - r_{21}u - r_{22}v), & (t, x) \in (0, +\infty) \times \Omega, \end{cases} \tag{6}$$

where all the coefficients are strictly positive. We endow the system with homogeneous boundary conditions.

Reference

[1] Murray, J. D., and Murray, J. D. (2003). Mathematical biology I: an introduction. New York: springer.
[2] Murray, J. D., and Murray, J. D. (2003). Mathematical biology II: spatial models and biomedical applications (Vol. 18, p. 811). New York: springer.